

Hourly-averaged solar plus wind power generation for Germany 2016: Long-term prediction, short-term forecasting, data mining and outlier analysis

David A. Wood

DWA Energy Limited, 25 Badgers Oak Bassingham, Lincoln, LN5 9JP United Kingdom

ARTICLE INFO

Keywords:

Country-wide renewable power generation
Combined solar and wind power planning
Predictions integrating diverse variables
Short-Term time series power forecasts
Prediction outlier analysis data filtering

ABSTRACT

Nationwide, hourly-averaged solar plus wind power generation (MW) data compiled for Germany for year 2016 is evaluated with ten influencing variables. Those variables cover, on an hourly basis, weather and ground-surface conditions and electricity prices. The transparent open box (TOB) algorithm accurately predicts and forecasts power generation (MW) for this dataset (prediction RMSE = 1175 MW and $R^2 = 0.9804$; hour ahead forecast RMSE = 1632 MW and $R^2 = 0.9609$) and meaningfully data mines the prediction outliers. Some 1.5 % of the data records display significant prediction errors. These records are mined to reveal that many of them form trends on a few specific days displaying unusual and rapidly changing weather conditions. Derivatives of ground level solar radiation, wind velocity and air pressure can meaningfully distinguish such unusual conditions and can be used to filter the dataset to further improve prediction accuracy. Derivatives and ratios of variables are also exploited to focus and modify feature selection for TOB analysis on approximately 10 % of the dataset (900 data records) responsible for the least accurate predictions. This more focused feature selection improves prediction accuracy for these more difficult to predict data records (RMSE improves from 3544 to 2630 MW; R^2 from 0.8027 to 0.8938).

1. Introduction

Many sustainable cities of the future intend to rely on renewable energy power supply. Solar and wind power generation supplies are likely to contribute significantly to such power supplies. However, both of these renewable energy sources are intermittent and vary significantly as short-term weather, medium term seasonally-related environmental conditions fluctuate. Moreover, market conditions cause day-ahead power prices to fluctuate influencing the commercial viability of wholesale customers purchasing such power supplies. This makes these important power sources notoriously difficult for system operators and power regulators to predict and plan for on a medium and long-term basis or forecast just a few hours ahead. A substantial amount of the research on methods to improve power prediction focuses on the performance of specific renewable-power-generation sites. However, there is a need for sustainable cities to consider collectively overall power supply on a much broader scale, i.e., from the wider region from which their power supplies are gathered, transmitted, ultimately distributed and sold to their mainly urban-located customers. This study considers the prediction and forecasting of solar and wind power generation on a country-wide basis for Germany. It does so with a dataset compiled on an hourly basis for 2016. Its findings are directly relevant to the challenges facing individual urban areas seeking to better plan

for, predict and forecast their future renewable energy power supplies.

Machine-learning predictions and forecasts of various types of renewable power generation using time-series of data variables (Antonanzas et al., 2016; Brown, Katz, & Murphy, 1984; Voyant et al., 2017) are essential for providing reliable expectations for the generation of power from short term (i.e. a few hours forward) to long term (weeks to months to years in the future). Such forecasts facilitate efficient balancing of electricity grids and planning for infrastructure capacity additions to distribute power more efficiently to the places it is needed when it is in demand. Machine-learning methods commonly applied for power time-series predictions and forecasts include neural networks (Mohandes, Rehmann, & Halawani, 1998), support vector regression (Chen, Liu, & Wu, 2011) linear, non-linear and boosted regression algorithms (Feng & Zhang, 2018), nearest neighbour (kNN) (Mangalova & Agafonov, 2014), kernel functions (Treiber, Heinemann, & Kramer, 2016) and various hybrid regression /fuzzy logic (Sivaneasan, Yu, & Goh, 2017), decision tree (Wang, Li, Ran, Che, & Zhou, 2018) and deep learning techniques (Gensler, Henze, Sick, & Raabe, 2016).

For wind and solar weather data in time series dominate the variables focused upon (Foley, Leahy, Marvuglia, & McKeogh, 2012; Rana, Koprinska, & Agelidis, 2016), with solar radiation (Inman, Pedro, & Coimbra, 2013) and wind velocities (Kusiak, Zheng, & Song, 2009)

E-mail address: dw@dwasolutions.com.

<https://doi.org/10.1016/j.scs.2020.102227>

Received 13 August 2019; Received in revised form 29 February 2020; Accepted 22 April 2020

Available online 05 June 2020

2210-6707/ © 2020 Elsevier Ltd. All rights reserved.

favoured, but other exogenous weather-related data contributing (Vladislavleva, Friedrich, Neumann, & Wagner, 2013). However, hourly country-wide renewable energy datasets involving diverse variables are not widely available (Stetco, Dinmohammadi, & Zhao, 2019).

An hourly-averaged dataset for 2016 that covers wind and solar power generation for the whole country of Germany has recently been compiled from published information (OPSD, 2019) that is suitable for evaluation using various machine learning techniques (Ferreira, 2018). Here, it is evaluated using an optimized data matching machine learning algorithm (Wood, 2019a). The data variables considered in that study are diverse in that they include power load, a range of weather-related metrics, conditions of the ground surface morphology and hourly traded power prices (from a day-ahead trading exchange index). The transparent open box (TOB) algorithm (2018b, Wood, 2018a, 2019b) has demonstrated its effectiveness for prediction and data mining for that wind plus solar country-wide data set (Wood, 2019a). This study expands on that evaluation to demonstrate how the optimized data matching algorithm can also provide useful short-term forward-looking power forecasts and more in-depth data mining of the outliers and prediction errors identified in that initial study (Wood, 2019a).

2. Country-wide renewable power dataset for Germany 2016

Open Power System Data (OPSD, 2019) publishes historic data for the European power system. 8784 data points for Germany 2016 (one for each hour in the year) make up the compiled dataset evaluated. These data combine two sets of OPSD information.

- **OPSD Time Series (2019)**, incorporating solar generation (MW) wind generation (MW) and price data (Euro/MWh; derived from Energinet, 2019);
- **OPSD Weather Series (2019)**, weather-related variables and environment-related variables derived from 256 weather stations positioned at strategic locations throughout Germany.

Pre-processing of the data to achieve a consistent hourly-averaged dataset consisting of eleven variables was required. This was facilitated using Visual Basic for Applications (VBA) code. The TOB algorithm is applied to this dataset using Excel and VBA code. The variables are statistically characterized in Table 1.

The weather-related variables (8 out of 10 independent variables) are significantly influenced by daily, regional and seasonal fluctuations. The environmental variable, surface roughness (Z0) is also influenced by seasonal and regional variations, i.e., in crop height and foliage density. An hourly day-ahead power price index (PrI), derived specifically for the TOB analysis, is also influenced by daily, seasonal and regional market factors. PrI is calculated as the hourly price less the mean annual day-ahead price, and that value divided by the mean of the annual day-ahead hourly prices recorded (Eq. 1).

$$PrI = (Prh - Prmean)/Prmean \quad (1)$$

Prh represents the hourly day-ahead price at any hourly point during the year;

Prmean represents the mean hourly day-ahead price averaged over the entire calendar year.

5012 (57 %) of the data records make non-zero MW contributions from solar generation (Wood, 2019c). Consequently, wind without additions from solar installations contributed power to 3772 of the hourly data records during 2016. Correlations between the total (wind plus solar) hourly power generated are stronger with wind velocity than solar radiation. The other input variables are not well correlated with combined wind and solar power generated (Fig. 1).

Fig. 1 highlights that the hourly data is not evenly distributed across the entire solar plus wind power generation (MW) range. There are many more data records in the < ~15,000 MW range and much fewer in the > ~30,000 MW range. This irregular data spread (Fig. 1) impacts the prediction reliability for data matching algorithms in certain sectors of the wind plus solar power generation distribution. TOB does not use correlations, meaning that it is able to meaningfully consider variables that display low correlations with MW for the power generation distribution as a whole but possess meaningful seasonal relationships (e.g. surface roughness, Fig. 1C). Such variables can and do make significant contributions to TOB's MW predictions. Wind speed, solar radiation and MW generated can vary significantly over periods of successive hours on some days (Fig. 1). Such rapid variations coupled with the broad dispersion of the input variables versus MW presents a significant prediction challenge.

3. TOB machine-learning method

TOB optimized data matching facilitates prediction and data-mining (Wood, 2018a, b; 2019b) and, as will be shown here, can also be configured for time-series forecasting. The majority of machine learning algorithms applied to renewable power forecasting use correlations, regressions and/or other statistical relationships. TOB does not seek to exploit such dependencies among its input variables. This enables it to consider dataset relationships in a quite distinct way from most machine-learning algorithms. Seeking patterns between multiple input variables across all available data records provides TOB with the ability to identify distinctive relationships in specific time sequences of data records, even when the variables are not well correlated with the dependent variable throughout the dataset. The TOB prediction methodology applied to time series of renewable data is described in detail elsewhere (Wood, 2019a). Its two-stage process (TOB Stages 1 and 2) is summarized here and expanded for time-series forecasting with the aid of a schematic (Fig. 2).

The method's transparency and resistance to over-fitting (Lever, Krywinski, & Altman, 2016) provide it with some advantages over correlation-based machine-learning algorithms. This is particularly the case when dealing with dispersed, unevenly distributed and poorly

Table 1
County-wide (Germany for calendar year 2016) hourly-averaged solar plus wind power generation data variables (OPSD data, 2019 compiled by Wood, 2019a).

Variable	Symbol	Units	Min	Max	Mean
Day-ahead price ratio	PrI	ratio	-5.489	2.622	0.000
Wind velocity at 1 m above displacement	V1	m/s	0.747	7.928	2.945
Wind velocity at 10 m above displacement	V10	m/s	1.346	11.164	4.298
Wind velocity at 50 m above displacement	V50	m/s	1.959	14.923	6.058
Top of atmosphere horizontal radiation	SWT	W/m ²	0.000	1160.939	277.797
Ground horizontal radiation	SWG	W/m ²	0.000	885.726	145.734
Surface roughness	Z0	m	0.141	0.235	0.200
Temperature at 2 m above displacement	TC	°C	-8.553	28.695	9.222
Air density	Rho	kg/m ³	1.134	1.307	1.212
Air pressure	Pa	Pa	95,468	100,957	98,551
Solar plus wind power generated (dependent variable)	MW	MW	374	43,860	12,702

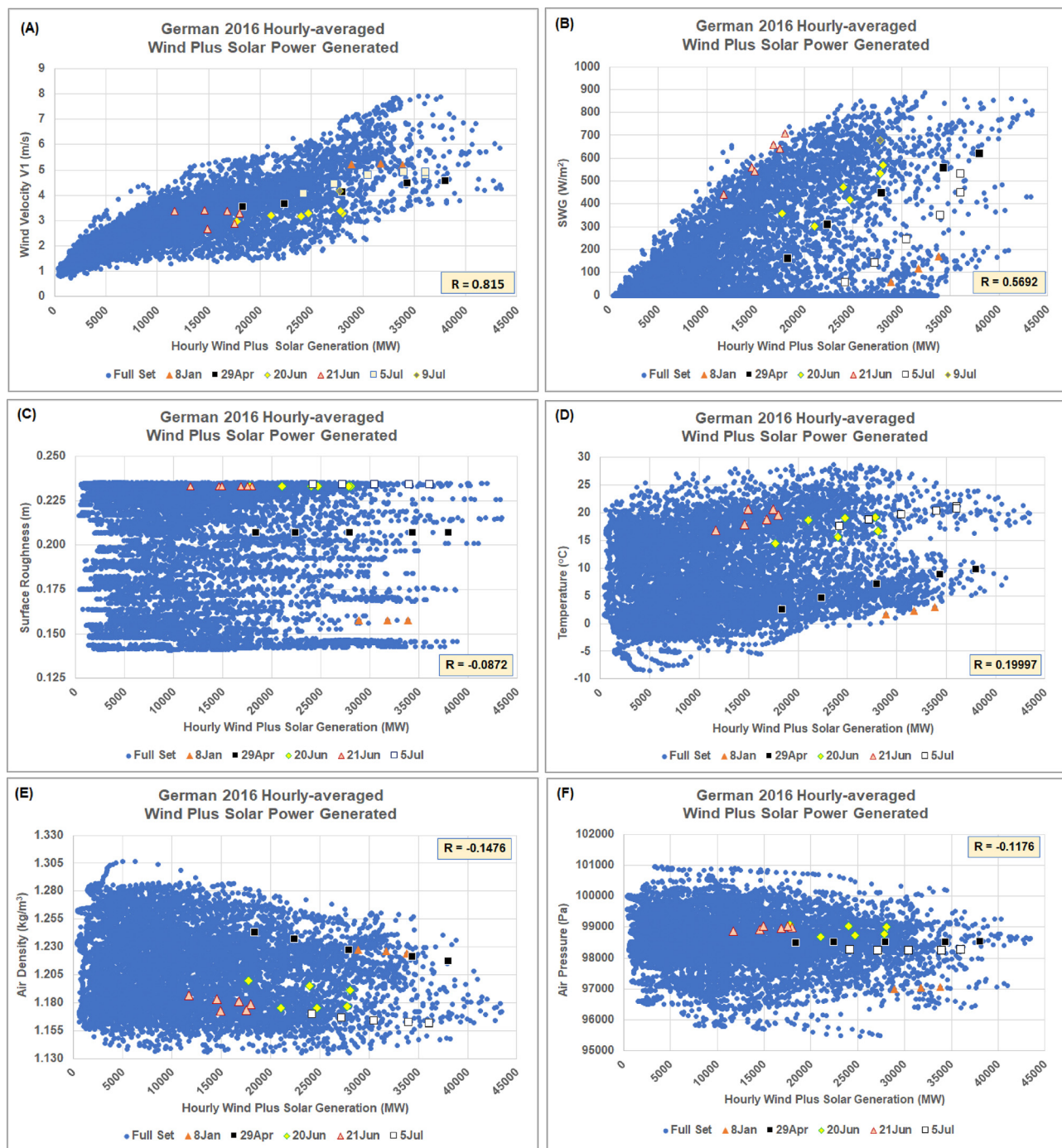


Fig. 1. The six input variables displayed contribute most to the TOB Stage 2 optimum prediction solution (Tables 2 and 3). R is correlation coefficient. Five hourly sequences identified for specific days are associated with rapid and significant MW changes over short periods of time. Each graph (A) to (F) includes 8784 hourly data records.

correlated variables typical of renewable power generation datasets. Similar benefits are also associated, to an extent, with lazy learning and locally-weighted methods (Atkeson, Moore, & Schaal, 1997; Birattari, Bontempi, & Bersini, 1999; Cover & Hart, 1967), as well as nearest-neighbour and k-learning approaches (Chen & Shah, 2018). However, TOB does not employ bagging, clustering and linearization to establish the best matching data records. It derives and sums, on an equally-weighted bases for TOB Stage 1, of the squared errors for each independent variable among the available data record in its search for the closest matching data records. Stage 1 typically establishes the top-ten matching records for each data record to be predicted or forecast in that

way. The weightings for the best matches are then optimized (Stage 2). The simple Stage 1 criteria (variable weights held constant; $Q = 10$) applied initially generates useful predictions and data mining insight. The results of the two stages considered together complement each other and help to verify statistical significance and identify cases of over fitting.

A customized memetic firefly optimizer has been developed for TOB, exploiting metaheuristic profiling and recording multiple good solutions that assist data mining (Wood, 2019). This uses an adaption of the firefly algorithm (Yang & He, 2013) enhanced with five other metaheuristics. For smaller tuning subsets Excel's standard Solver

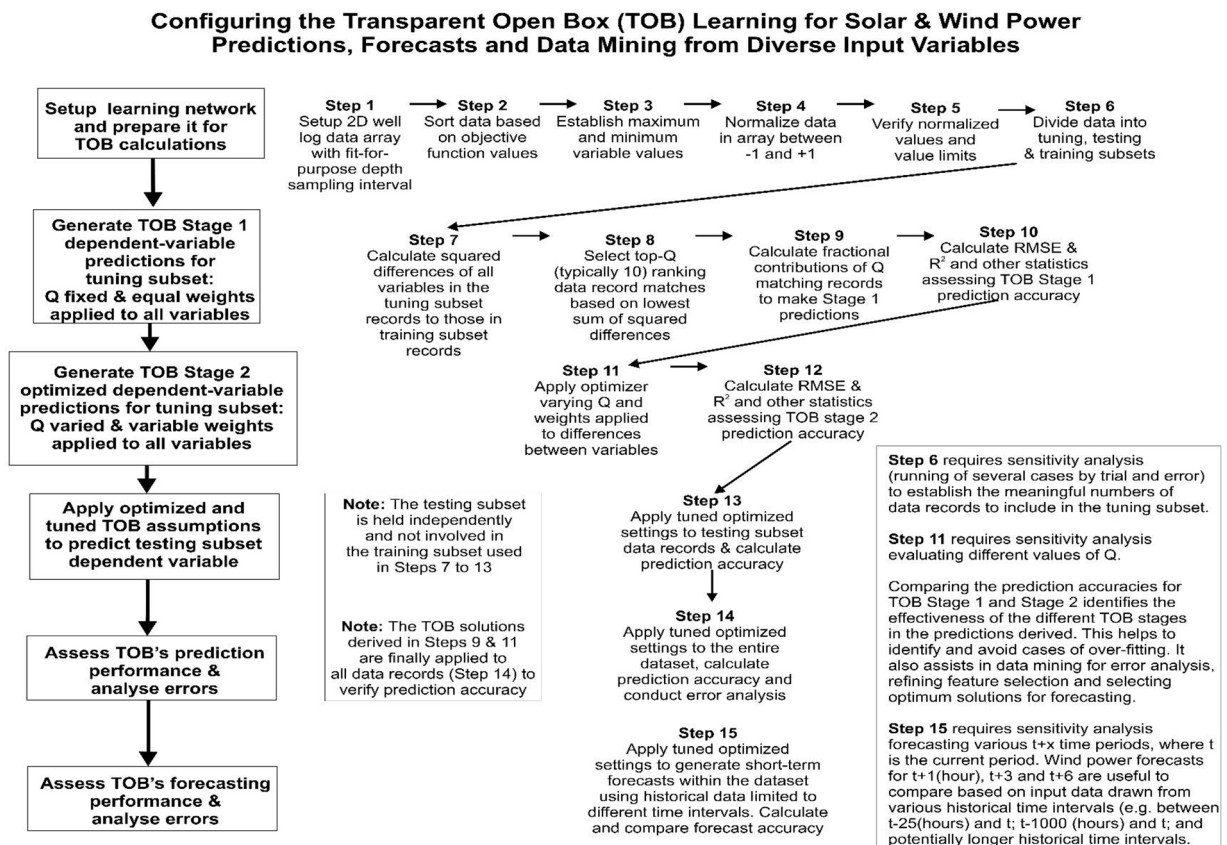


Fig. 2. Fifteen steps are involved in conducting TOB prediction tuning, time-series forecasting and error / outlier data mining for renewable power datasets as shown in this schematic.

optimizers (Generalized Reduced Gradient (GRG) and Evolutionary - Frontline Solvers, 2019) are also exploited by TOB. The optimization performed for this study is coded in VBA.

4. Prediction, forecasting and data mining results

4.1. Tuning TOB predictions for renewable power generation time series

Five cases, each with different small tuning subsets (147 records), were evaluated for prediction tuning using the dataset described (Wood, 2019a). For each case the TOB Stage 2 was optimized independently using three optimizers leading to fifteen distinct optimized prediction solutions. These solutions were evaluated initially using small independent testing subsets (197 records) and ultimately against all dataset records for calendar year 2016. The wind plus solar MW

tuned prediction results are summarized in Table 2.

The accuracy of the TOB Stage 2 optimum tuned solutions applied to the entire 2016 data set are impressive: RMSE ranging from 1174 to 1310 MW; R² ranging from 0.9755 to 0.9804 for the fifteen solutions derived. The prediction accuracies of individual tuning subset solutions depend, to an extent, on whether their tuning subsets sample more or less of the more unusual (outlier) data records that are difficult to fit. It is therefore essential to run several cases to verify reproducibility. The close agreement between the testing subset and full dataset prediction accuracies of the solutions generated is reassuring (Table 2). It verifies the validity and reproducibility, collectively, of the tuning subset solutions generated. It thereby justifies employing tuning subsets sampling relatively few data records to derive the optimization solutions. The recorded improvement in prediction accuracy of the TOB Stage 2 solutions, compared to the TOB Stage 1 solutions, when applied to all

Table 2

Independent testing subset and entire dataset prediction accuracies achieved by the TOB algorithm for Germany country-wide 2016 solar and wind hourly power generation data. RMSE = root mean squared error; AAPD = average absolute percentage error, SD = standard deviation, R² = coefficient of determination (formulas used for these statistical measures of accuracy are defined in Wood, 2019a).

TOB Prediction Solutions	Statistical Measures of Prediction Accuracy			
	RMSE	AAPD%	SD	R ²
Testing Subset (197 data records)				
Mean for 5 Stage 1 solutions derived	1411.6	9.2063	1414.1	0.9732
Mean for 15 Stage 2 solutions derived	1282.6	8.1789	1283.2	0.9777
Best case Stage 1	1203.2	8.6184	1206.0	0.9812
Best case Stage 2	1070.5	6.9106	1073.0	0.9840
Full Dataset (8784 data records)				
Single Stage 1 solution	1388.9	9.3264	1388.5	0.9726
Mean for 15 Stage 2 solutions applied	1230.3	8.2485	1229.9	0.9784
Best case Stage 2	1174.6	7.3702	1173.5	0.9804

Table 3

TOB optimum solution for the best case in terms of prediction performance accuracy for the full data set. Variable symbols are defined in Table 1.

Variable	TOB Stage 1	TOB Stage 2
Q	10	9
Variable error weights applied		
Prl	0.5	0.000272124
V1	0.5	0.020886479
V10	0.5	0.000418697
V50	0.5	0.000033827
SWT	0.5	0
SWG	0.5	0.001159923
Z0	0.5	0.998212695
TC	0.5	0.028837704
Rho	0.5	0.002650696
Pa	0.5	0.040868087

the data records available rules out overfitting.

Table 3 provides the TOB solution details for the best optimized solution among the 15 solutions derived (Wood, 2019a). The best case TOB Stage 2 solution assigns its highest variable error weight to surface roughness (Z0). Temperature (TC), wind velocity at 1 m (V1) and atmospheric pressure (Pa) are also assigned higher variable error weights than other variables by the best optimized solution. All variables, except horizontal solar radiation at the top of the atmosphere (SWT) are assigned a non-zero weight and therefore contribute to the predictions. In other cases, the input variable weights selected by TOB Stage 2 differed, but generally Z0 was assigned the highest weight, horizontal solar radiation at ground level (SWG) was favoured over SWT and V1 over V10 and V50. The day-ahead price index (Prl) was assigned a low non-zero weight in fourteen of the fifteen solutions derived.

Fig. 3 displays predicted versus actual MW values for all 8784 data records for the best TOB Stage 2 solution found. Although prediction accuracy is high there are a few significant outlier predictions. Two of the more extreme outliers (data records #4118 for 20/06/2016 13:00:00 and #4138 for 21/06/2016 09:00:00 less than 24 h later) are identified in Fig. 3, outlying on opposite sides of the main MW prediction trend.

Segmental analysis of MW prediction distribution reveals that the poorest prediction accuracies are associated with the middle range between ~10,000 to ~30,000 MW. The highest MW prediction accuracy is associated with the lower generation range (< 10,000 MW) which contains about 50 % of the data records. Outlier analysis is therefore focused on the wind plus solar power generation range > 10,000 MW.

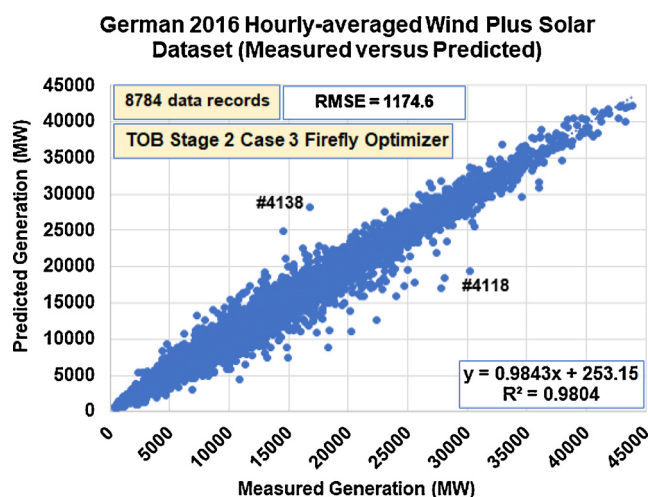


Fig. 3. German 2016 country-wide hourly TOB predicted solar plus wind power generated (MW). Data for 8784 data records displayed.

Table 4

TOB prediction components for an example data record. Hourly records are numbered sequentially from 1 to 8784 h in German 2016 wind plus solar power data set.

Typical Prediction (as per > 98 % of Records)		
Data Record Predicted : 1693 (11/03/2016 12:00)		
Best Matching Records	Fractional Contribution	Normalized Component of MW Prediction
TOB Stage 1 Optimized Prediction (Q = 10)		
1692	0.2684	−0.0970
1691	0.2390	−0.0872
1717	0.0959	−0.0481
1716	0.0664	−0.0301
1715	0.0637	−0.0281
1766	0.0610	−0.0065
1765	0.0572	0.0002
1690	0.0568	−0.0260
1694	0.0535	−0.0269
1714	0.0382	−0.0189
Normalized Stage 1 Prediction:		−0.3686
Predicted MW TOB Stage 1:		14,102
Measured MW #1693:		13,179
Error (Measured - Predicted):		−923
TOB Stage 2 Optimized Prediction (Q = 9)		
1692	0.4834	−0.1746
1694	0.3314	−0.1669
1691	0.1016	−0.0371
1690	0.0310	−0.0142
1717	0.0164	−0.0082
1716	0.0143	−0.0065
1715	0.0122	−0.0054
1766	0.0051	−0.0005
1765	0.0046	0.0000
1714	0	0
Normalized Stage 2 Prediction:		−0.4134
Predicted MW TOB Stage 2:		13,128
Measured MW #1693:		13,179
Error (Measured - Predicted):		51

4.2. Data mining individual predictions

The TOB network provides easy access to the ten best matches (i.e., those with the lowest combined squared errors between their variables) for each data record prediction or forecast. It also provides details of their exact percentage contributions to individual predictions and forecasts. This is valuable information for data mining and is the basis of the algorithm's transparency. Table 4 illustrates this for a typical data record that yield's accurate TOB prediction. Some of the closest-matching records identified by TOB Stage 1 are commonly adjacent records to the record being matched. This is particularly so during stable weather periods. This is the case for record #1693 and the TOB Stage 2 solution increases the relative contribution of the adjacent records to the prediction, improving its accuracy.

However, stable conditions do not prevail throughout the year and during unsettled and/or rapidly changing and/or regionally contrasting weather conditions data records can match with records from quite distinct periods. In such cases, the quality of the MW predictions depend on how densely populated the MW data variable distributions are in the vicinity of the record being matched. Many of the extreme prediction outliers are associated with abnormal and rapidly changing weather conditions.

4.3. Short-term country-wide solar plus wind power forecasts using the TOB Algorithm

The TOB Stage 2 tuned solutions for based on all hourly records in the 2016 dataset (8784 data records) are usefully applied to provide short-term power predictions, e.g. one or more hours ahead using only historic data. Here, 7784 hourly records (from record #1001 onward through to record #8784) are forecast using different durations of backward-looking data (excluding any current or future data with respect to each data record). Three historic time durations with respect to each current record (designated as time t) are considered beginning one hour, three hours and six hours before time t . These provide power forecasts at:

- A $t + 1$ (1 h ahead of period t)
- B $t + 3$ (3 h ahead of period t)
- C $t + 6$ (6 h ahead of period t)

Five different time ranges of historic data records earlier in the year 2016 relative to time t are considered for each of scenarios A, B and C:

- $t-25$ to t (the most recent 25 -h period before t); about one day before t
- $t-50$ to t (the most recent 50 -h period before t); about two days before t
- $t-100$ to t (the most recent 100 -hs period before t); about four days before t
- $t-200$ to t (the most recent 200 -hs period before t); about eight days before t
- $t-1000$ to t (the most recent 1000 -hs period before t); about forty-two days before t

The first 1000 data records of the 2016 data set are excluded from the forecasting analysis because they are included to varying extents in the forecasting from data records #1000 working forward hour-by-hour through the dataset.

Accuracies of the forecasts for the scenarios described, applying TOB Stage 1 followed by the optimum TOB stage 2 solution derived from analysis of the full-year dataset are shown in achieved for these three short-term forecasting scenarios are listed in Table 5.

The RMSE for each scenario and time period evaluated for the forecasts are compared in Fig. 4 for each of the historical range of data records involved.

As to be expected, $t + 1$ forecasts are the most accurate and $t + 6$ forecasts are the least accurate (Table 5). Likewise, the accuracy of the forecasts improve as the number of prior hours included for data matching increases. The forecasting accuracies achieved for $t + 1$ are inferior to those derived in tuning TOB with the full year of hourly data records (Table 2) but remain credible and useful forecasts..

With only 25 h of prior data available for forecasting, the forecasts for period $t + 1$ (TOB Stage 2 RMSE = 1942.1 MW; R^2 = 0.9446; Table 5) are the poorest derived. However, the forecasting accuracy progressively improves for TOB Stage 1 and Stage 2 forecasts as more prior data becomes available for data matching. With 1000 h of prior data available, TOB Stage 2 achieves forecasting accuracy of RMSE = 1631.5 MW; R^2 = 0.9609 (Table 5). Some additional improvements in accuracy for $t + 1$ forecasts could reasonably be expected by increasing the duration of the prior time period beyond 1000 h for which potentially matching data records are made available.

The $t + 3$ forecasts, although clearly inferior to the $t + 1$ forecasts for both TOB Stage 1 and Stage 2 analysis (Table 5), still provide credible accuracy. For 1000 h of prior data available, TOB Stage 1 achieves forecasting accuracy of RMSE = 2421.4 MW; R^2 = 0.9150 (Table 5), and TOB Stage 2 is unable to improve upon that. For short periods of prior data available the forecast accuracy for $t + 3$ is much reduced (e.g., for 25 h of historical data available TOB Stage 2

Table 5

Short-term Country-wide Wind Plus Solar Power Forecast Accuracy for Germany 2016 for 7784 Hourly Data Records Assessed Individually Using Near-past Historical Data Records of Time Intervals of 25–1000 Hours Prior to Time t

Short-term Country-wide Wind Plus Solar Power Forecast Accuracy for Germany 2016 for 7784 Hourly Data Records Assessed Individually Using Near-past Historical Data Records of Time Intervals of 25–1000 Hours Prior to Time t															
	TOB Stage 1						TOB Stage 2								
	RMSE	MSE	APD% AAPD%			R	R ²	RMSE	MSE	APD% AAPD%			SD	R	R ²
Historical Sequences used to match for forecast $t + 1$ h															
$t-25$ to t	2082.6	4.3E+06	-4.0	15.8	2080.7	0.9677	0.9364	1942.1	3.8E+06	-3.3	14.7	1942.0	0.9719	0.9446	
$t-50$ to t	1989.6	4.0E+06	-3.6	14.8	1986.7	0.9707	0.9422	1772.5	3.1E+06	-2.9	13.4	1771.9	0.9767	0.9538	
$t-100$ to t	1936.2	3.7E+06	-3.1	14.2	1930.7	0.9725	0.9457	1700.9	2.9E+06	-2.5	12.7	1699.4	0.9786	0.9576	
$t-200$ to t	1867.0	3.5E+06	-3.1	13.8	1862.0	0.9744	0.9495	1651.3	2.7E+06	-2.1	12.3	1649.7	0.9798	0.9600	
$t-1000$ to t	1798.5	3.2E+06	-2.9	13.0	1793.8	0.9763	0.9531	1631.5	2.7E+06	-2.0	11.9	1629.9	0.9803	0.9609	
Historical Sequences used to match for forecast $t + 3$ h															
$t-25$ to t	3494.3	1.2E+07	-8.7	28.9	3481.2	0.9067	0.8221	3410.6	1.2E+07	-7.5	28.2	3390.8	0.9116	0.8311	
$t-50$ to t	3111.0	9.7E+06	-7.7	25.2	3102.1	0.9266	0.8586	3006.0	9.0E+06	-7.2	24.6	2998.1	0.9316	0.8679	
$t-100$ to t	2905.5	8.4E+06	-6.4	23.1	2891.4	0.9368	0.8776	2826.6	8.0E+06	-6.0	22.6	2815.5	0.9400	0.8837	
$t-200$ to t	2678.5	7.2E+06	-6.0	21.6	2667.8	0.9465	0.8958	2655.7	7.1E+06	-5.4	21.2	2646.9	0.9472	0.8971	
$t-1000$ to t	2421.4	5.9E+06	-5.0	19.1	2412.2	0.9566	0.9150	2492.5	6.2E+06	-5.0	19.5	2485.8	0.9536	0.9093	
Historical Sequences used to match for forecast $t + 6$ h															
$t-25$ to t	4469.6	2.0E+07	-13.9	40.0	4444.5	0.8442	0.7126	4501.3	2.0E+07	-11.8	39.5	4433.3	0.8434	0.7114	
$t-50$ to t	3777.1	1.4E+07	-11.9	33.1	3765.4	0.8899	0.7919	3717.2	1.4E+07	-11.3	32.8	3703.3	0.8936	0.7986	
$t-100$ to t	3404.4	1.2E+07	-9.2	28.9	3384.8	0.9120	0.8318	3395.6	1.2E+07	-9.0	28.9	3377.4	0.9124	0.8324	
$t-200$ to t	3052.4	9.3E+06	-8.2	26.1	3037.4	0.9299	0.8647	3095.7	9.6E+06	-8.0	26.3	3082.9	0.9276	0.8604	
$t-1000$ to t	2635.3	6.9E+06	-6.1	21.8	2623.3	0.9484	0.8994	2786.4	7.8E+06	-6.7	23.1	2777.2	0.9417	0.8868	

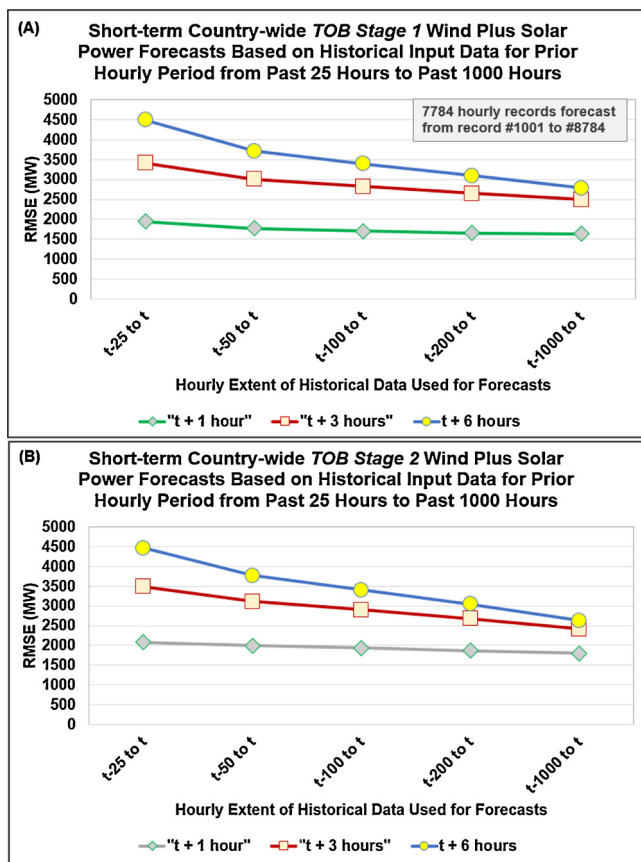


Fig. 4. $t+1$, $t+3$ and $t+6$ forecasts compared for various ranges of prior hours used by TOB analysis. (A) TOB Stage 1 solutions; (B) TOB Stage 2 solutions.

RMSE = 3410.6 MW; $R^2 = 0.8311$; Table 5). For the $t+3$ analysis, as the period of prior data available increases the forecast accuracy increases significantly (Fig. 4) for both TOB Stage 1 and Stage 2. The same is true for $t+6$ forecasts, the accuracy of which, although significantly inferior to that of $t+3$ forecasts, improves at an even faster rate than the shorter-term forecasts (Fig. 4). These results suggest that by using much longer prior data available periods (e.g., 365 days or perhaps even historical data from multiple prior years), the TOB Stage 1 data matching process should lead to additional improvements in accuracy for the $t+3$ and $t+6$ forecasts.

It is to be expected that the TOB algorithm, with relatively short prior data available for TOB Stage 1 data matching, result in less accurate $t+6$ forecasts. This is due to many data records available for TOB prediction tuning tend to be those immediately adjacent to the records being predicted (i.e., one or two hours before or after that period). However, for $t+6$ forecasts, the five hourly data records in the time series immediately prior to the record being forecast (and all periods in the dataset occurring after $t+6$) are not available for TOB matching. However, it is likely that, by involving longer historical time periods, $t+6$ forecasts (and even further ahead medium-term forecasts) can find close matches across the previous 12 months that could potentially rival those unavailable adjacent records. For historical data periods of 200 and 1000 h the TOB Stage 1 solution actual provides slightly more accurate $t+6$ forecasts than TOB Stage 2 (Table 5). This occurs because the TOB Stage 2 solutions that are optimized for prediction purposes are based on an extensive training subset that includes many data records immediately adjacent in the time series to each data record being predicted. Such solutions are clearly insufficiently tuned to optimize forecasts when the five data records immediately preceding each data record to be forecast are not available for data matching (i.e.,

the $t+6$ forecast scenario). In effect, that TOB stage 2 tuned solution is attempting to overfit the more limited data available to it. This problem may be overcome if more prior data records are made available for forecasting (i.e., > 1000 h), but further analysis is required to verify this. Also, it may be possible to alternatively tune the TOB Stage 2 solution exclude data from surrounding hours. More analysis is required on data sets covering multiple years to address such issues.

The forecasting analysis presented here demonstrates that credible accuracies for short-term time-series power forecasting can be achieved using the TOB algorithm. The potential to apply TOB for medium-term and long-term forecasting requires further studies.

5. Prediction outlier analysis

5.1. Identifying and explaining extreme prediction outliers

To segregate the prediction outliers from the more accurate predictions made during the TOB tuning process, a series of filters are applied to the TOB MW prediction versus recorded measurements. A filter applying a squared prediction error (SE) of 2×10^7 (plus or minus prediction error of $> = 4472$ MW) to TOB Stage 1 identifies about 1.5 % of the data records (129 in total) with prediction errors above that value. An SE filter of 1×10^7 (plus or minus prediction error of $> = 3162$ MW) identifies about 4.2 % of the data records (373 in total) with prediction errors above that value. Whereas, an SE filter of 5×10^6 (plus or minus prediction error of $> = 2236$ MW) identifies about 10.2 % of the data records (900 in total) with prediction errors above that value. It is possible to identify certain hourly sequences, on a few specific days in the 2016 dataset, that account for many of the larger prediction errors. These hourly sequences are related to unusual and rapidly changing weather (i.e., ground level solar radiation and wind velocity) conditions (e.g., 8th January, 29th April, 20th and 21st June and 5th July, Wood, 2019a). The hourly sequence of recorded wind plus solar power for these rapidly changing conditions are highlighted in Fig. 2. These hourly sequences represent some of the most challenging to predict or forecast accurately.

The 2016 dataset is averaged on a country-wide basis for Germany as a whole. This means that some significant regional variations in weather conditions can be smoothed out in certain hours. Storms and high cloud cover can affect certain regions of the country but not others. The power generation facilities (solar plus wind) in certain regions can therefore, on a few occasions, skew the nationally averaged values. Regional outages of substantial generation capacity could also lead to anomalous nationally averaged data. The contrasting county-wide average conditions recorded for 20th June and 21st June might be explained in such terms. However, detailed analysis of regional weather and grid availability data would be required to confirm this.

The majority of extreme TOB prediction and forecast outliers are a consequence of a lack of other data records in the dataset with a set of variable values comparable to those resulting from unusual and/or rapidly changing conditions. This makes it difficult for the data-matching process to find closely matching data records and can result in poor data record matches being selected as the best available. Such prediction outliers are mainly associated with hourly sequences on a few days experiencing complex and rapidly changing weather conditions. As those sequence plot as trends for certain variables, e.g. SWG versus MW (Fig. 2) it is possible to identify and isolate them for special analysis. Fig. 5 illustrates the impact on wind plus solar MW prediction performance of the best optimized TOB solution (Tables 2 and 3) by excluding the 129 most extreme outlier records. Future studies involving hourly data for multiple years will be conducted once the dataset can be expanded as data is released for years after 2016.

Fig. 6 illustrates the impact on wind plus solar MW prediction performance of the best optimized TOB solution (Tables 2 and 3) by excluding the 900 outlier records by applying an SE filter of 5×10^6 (plus or minus prediction error of $> = 2236$ MW). These 900 outliers

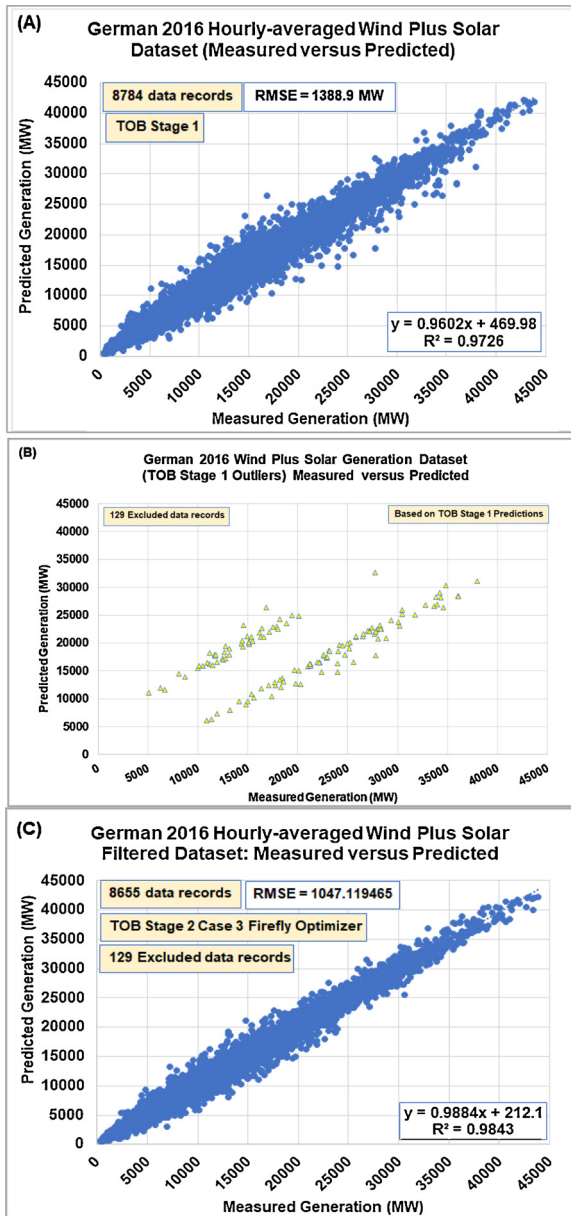


Fig. 5. A) Accuracy for best TOB Stage 1 solution unfiltered; B) 129 extreme prediction outliers excluded by the SE filter of 2×10^7 ; C) TOB Stage 2 accuracy with extreme outlier predictions excluded.

are analysed in section 5.3 to identify potential ways of improving their MW predictions.

Collectively, the 900 outlying predictions, in isolation (10.2 % of the complete dataset), only achieve a prediction accuracy of $RMSE = 3544$ MW and $R^2 = 0.8027$ applying the TOB Stage 1 solution derived for the entire. This compares with a prediction accuracy for the remaining 7884 data records (89.8 % of the complete dataset) of $RMSE = 846$ MW and $R^2 = 0.9895$ applying the TOB Stage 1 solution.

5.2. Association of prediction outliers with periods of rapidly changing weather conditions

The periods that tend to generate anomalous TOB MW predictions are associated, for the most part, with rapidly changing weather conditions often with sequential hours following observable trends. Consequently, it is possible to distinguish them and treat them separately for prediction purposes with more focused feature selection for

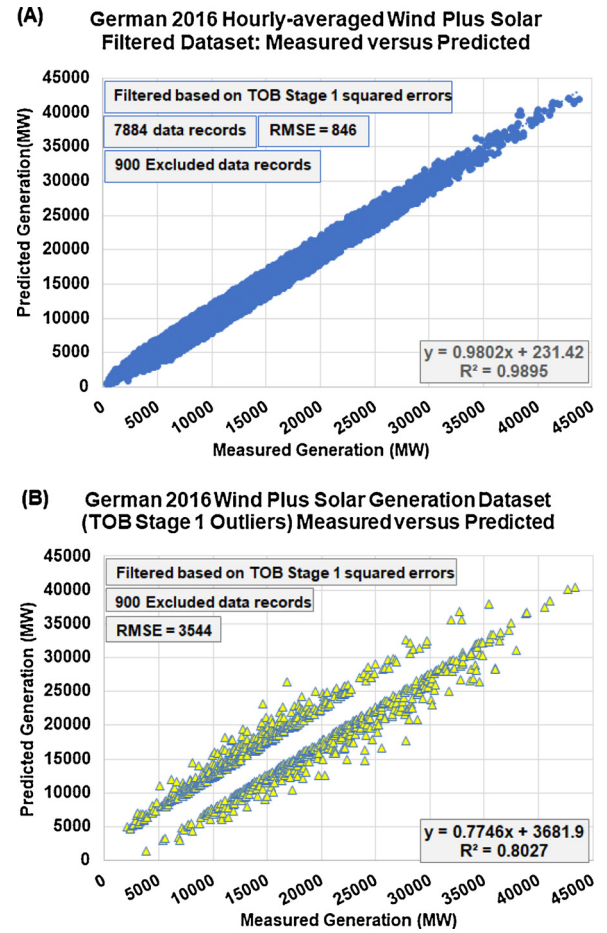


Fig. 6. A) accuracy for the TOB Stage 1 solution applied to the entire filtered dataset with 900 outlying predictions removed; B) The 900 prediction outliers excluded by the SE filter of 5×10^6 .

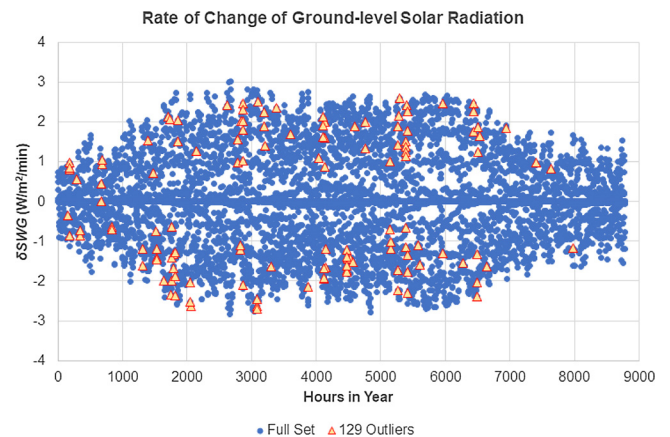


Fig. 7. Derivative (per minute rate of change) of horizontal ground-level solar radiation (δSWG) hourly averaged values for all 8784 data records with the 129 extreme prediction outliers distinguished by the SE filter of 2×10^7 .

the TOB learning network. Using ratios between some of the independent dataset variables and calculating derivatives of those variables offer potential ways to achieve this.

Derivatives of the hourly data are calculated by taking the differences between successive hourly data and dividing that difference by 60 to provide an average per minute rate of change for those variables most prone to rapid weather changes. The formulations for the derivative calculations for SWG, SWT and V1 are

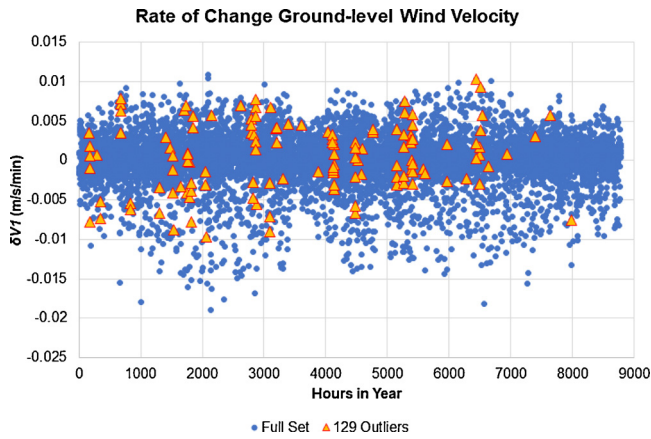


Fig. 8. Derivative ground-level wind velocity ($\delta V1$) hourly averaged values for all 8784 data records with the 129 extreme prediction outliers distinguished by the SE filter of 2×10^7 .

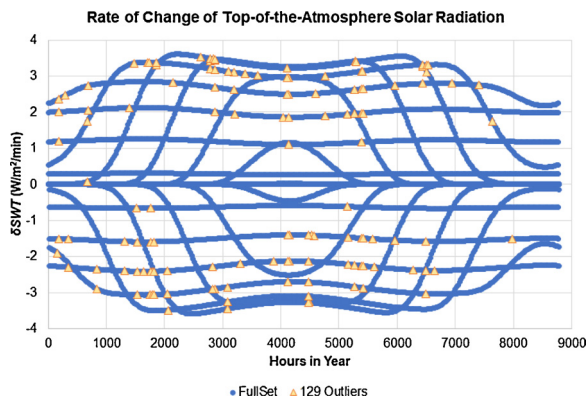


Fig. 9. Derivative (per minute rate of change) of horizontal top-of-the-atmosphere solar radiation (δSWT) hourly averaged values for all 8784 data records with the 129 extreme prediction outliers distinguished by the SE filter of 2×10^7 .

$$\delta SWG = (SWG_{(t=x)} - SWG_{(t=x-1)})/60 \quad (2)$$

$$\delta SWT = (SWT_{(t=x)} - SWT_{(t=x-1)})/60 \quad (3)$$

$$\delta V1 = (V1_{(t=x)} - V1_{(t=x-1)})/60 \quad (4)$$

where

δ refers to the derivatives or per-minute rates of change of the input variables

$(t = x)$ refers to the hourly averaged value of the variable at hour x during the annual hourly sequence; and,

$(t = x-1)$ refers to the hourly averaged value of the variable at hour $x-1$ during the annual hourly sequence (i.e., the hour immediately preceding hour x in the sequence).

Applying derivatives calculated in that way for solar radiation (δSWG and δSWT) and wind speed ($\delta V1$) variables for the complete dataset (8784 data records) makes it possible to distinguish most of the 129 extreme outliers (Figs. 7 and 8). Clearly, those graphs highlight that the outlying predictions are frequently associated with the periods experiencing most rapid change in SWG (Fig. 7) and V1 (Fig. 8) variables. Armed with such awareness, it is possible to use such derivatives to screen for those periods, typically sequences of just a few hours, that are most likely to generate anomalous predictions.

Fig. 9 plots the derivative δSWT value and reveals that all but 7 of the extreme prediction outliers are associated with periods of more rapidly changing SWT (i.e. $> +1.5$ or < -1.5 W/m²/min). This indicates that such prediction outliers are most likely to occur during the predictable periods of rapidly changing solar radiation. Such periods are linked more fundamentally to annual climatic features influenced by the Earth's specific orbit around the sun. For the 2016 German dataset, the more extreme prediction anomalies (triangles in Fig. 9) are concentrated in calendar-year quarters one to three. This is a feature that is worthy of further evaluation in future studies incorporating data from multiple years. It is possible that variables such as surface roughness display a more definitive range in the quarter four, or it may be simply an aberration of weather patterns in 2016.

5.3. Improving solar plus wind power predictions for periods of rapidly fluctuating weather conditions

A further TOB analysis was conducted to focus on just the 900 outliers identified with prediction errors of greater than or equal to 2236 MW (i.e., filtering the complete dataset with an SE filter of 2×10^7). These outliers (10.2 % of the complete dataset) are also mostly associated with high derivative solar radiation values. This focused 900-outlier-record TOB analysis substitutes some of the independent variables used in the TOB best solution presented in Tables 2 and 3 and Fig. 3 for ratios and derivatives of those variables. From the original full-dataset TOB Stage 1 analysis the RMSE value for these 900 outliers is 3544 MW (Fig. 6) and for the 129 extreme outliers RMSE = 5792 MW. The objective of the further TOB analysis is to see whether the high errors associated with the original TOB best solution applied to the full dataset can be reduced for the 900 outlying predictions by including selected variable ratios and derivative values in the model's feature selection.

The relationships between variables, their ratios and derivatives used in this focused 900-outlier TOB analysis are displayed in a correlation matrix (Table 6). As well as identifying significant correlations between PrI, V10, SWG, SWG/SWT and δPa and MW, Table 5 identifies significant correlations between SWG and V50/V10 and δSWG and $\delta V10$ for this problematic difficult-to-predict, subset of hourly records.

Table 6

Correlation matrix for variables used for the TOB analysis focused on the 900 outlier predictions from the full dataset (Fig. 6B). These are those data records segregated using a filter of $SE > = 5 \times 10^6$.

R	PrI	V10	SWG	Pa	V50/ V10	SWG/ T°C	SWG/ SWT	Deriv $\delta V10$	Deriv δSWG	Deriv δPa	MW
PrI	1	-0.3251	-0.2903	0.1494	0.0865	0.0356	-0.2139	-0.0256	0.0821	0.2305	-0.5180
V10		1	-0.1508	-0.3010	0.0795	0.0259	-0.2159	0.1015	-0.0064	-0.0969	0.6283
SWG			1	0.1642	-0.6307	0.0065	0.8097	0.1872	0.1129	-0.2377	0.4594
Pa				1	-0.1429	0.0547	0.3314	0.0569	0.0470	0.0259	-0.0933
V50/ V10					1	-0.0720	-0.5945	-0.3565	-0.3379	0.0857	-0.1833
SWG/T°C						1	0.0406	0.0424	0.0550	-0.0075	0.0347
SWG/ SWT							1	0.0759	0.0976	-0.2398	0.3435
Deriv $\delta V10$								1	0.6424	-0.0417	0.1366
Deriv δSWG									1	0.1993	0.0186
Deriv δPa										1	-0.3440
MW											1

Table 7

TOB statistical measures of prediction accuracy for the outlier subset (900 records).

TOB Prediction Solutions	Statistical Measures of Prediction Accuracy			
	RMSE	AAPD%	SD	R ²
Testing Subset (101 data records)				
Mean for 5 Stage 1 solutions derived	3479.2	21.5690	3476.0	0.8417
Mean for 15 Stage 2 solutions derived	2991.8	18.2543	2989.9	0.8741
Best case Stage 1	2785.1	17.2837	2792.4	0.8990
Best case Stage 2	2561.6	15.7619	2519.1	0.9079
Full Dataset (900 data records)				
Single Stage 1 solution	3278.3	20.7801	3278.7	0.8425
Mean for 15 Stage 2 solutions applied	2696.0	16.4145	2695.8	0.8878
Best case Stage 2	2629.6	16.1375	2628.3	0.8938

Table 8TOB optimum best-case solution for the 900-record outlier subset. Variable symbols are defined in Table 1. “ δ ” refers to derivatives of those variables (see text for definition).

Variable	TOB Stage 1	TOB Stage 2
Q	10	7
Variable error weights applied		
PrI	0.5	0.258222089
V10	0.5	0.101820029
SWG	0.5	0
Pa	0.5	0.063570157
V50N10	0.5	0
SWG/TC	0.5	0.994526981
SWG/SWT	0.5	0.013104109
δ V10	0.5	0
δ SWG	0.5	0
δ Pa	0.5	0.026286563

Table 7 shows the prediction accuracy achieved by the outlier TOB analysis. Table 8 provides details of the best optimized solution found (based on fifteen distinct runs of the TOB Stage 2 optimizers). This analysis does improve upon the prediction accuracy for these outliers as a group. RMSE is reduced from 3544 MW for TOB Stage 1 applying the original feature selection (Table 3) to 2630 MW for the feature selection of the more focused TOB Stage 2 solution (Table 8). Moreover, R² is increased from 0.8027 for TOB Stage 1 applying the original feature selection (Table 3) to 0.8938 for the feature selection used for the more focused TOB Stage 2 solution (Table 8). Fig. 10 displays cross plots of predicted versus actual MW analysis focused exclusively on the 900 outliers. These predictions compare very favourably with those displayed in Fig. 5B for these data records applying the original TOB feature selection and analysis.

Indeed, for more than 400 of those data records achieve prediction errors of RMSE less than plus or minus 2236 MW in this focused TOB analysis involving some ratios and variable derivatives. Whereas, none of these 900 data records were predicted with accuracy of less than plus or minus 2236 MW in the original TOB analysis. Although this represents a meaningful prediction accuracy improvement, the focused TOB analysis does not completely resolve the specific predictions for the 150 extreme outliers that persist with high prediction errors and are identified with the SE filter of 2×10^7 . For example data records #2827, #8068, #8513 and #3082, highlighted in Fig. 10 represent the most extreme under-estimates of MW by the outlier TOB analysis. The details of these four data records are considered in more detail.

The variables assigned the most significant weight in the TOB Stage 2 analysis (Table 8) are SWG/TC, PrI and V10. Pa, δ Pa and SWG/SWT are also assigned non-zero weights. These variable priorities are representative of the other optimized TOB solutions. The fifteen Stage 2 optimized solutions derived for the 900-outlier subset all assign non-zero weights to PrI, V10, SWG/SWT, Pa and δ Pa variables. This suggests that the subset of data records, characterized by rapidly changing

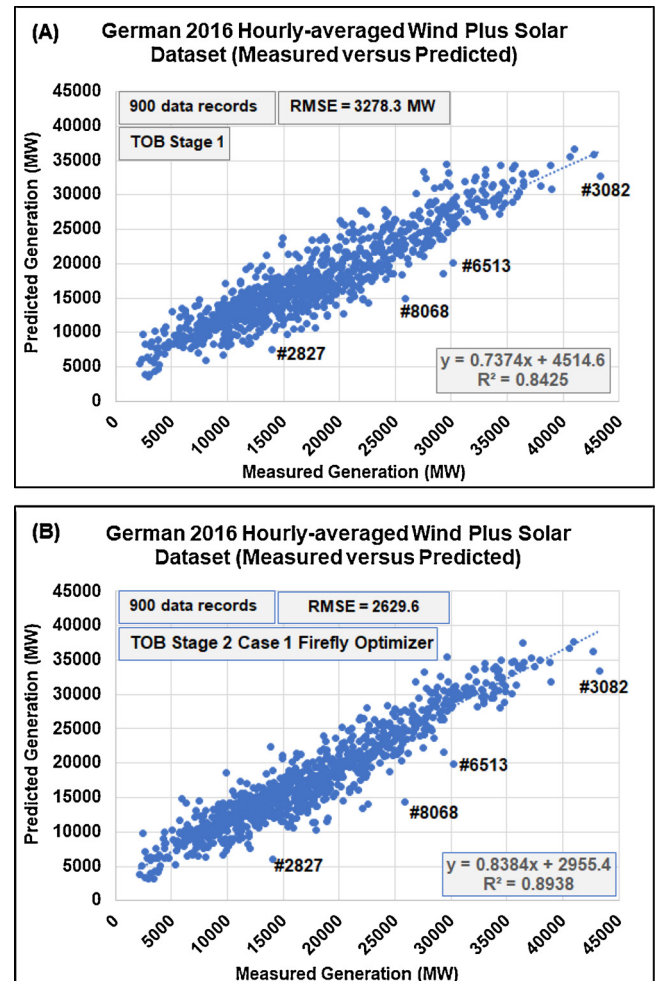


Fig. 10. A) Accuracy for the Stage 1 analysis (Table 7) applied to 900 outlier predictions from the original analysis (Fig. 6B); B) Accuracy for the Stage 2 analysis (Table 7) applied to 900 outlier predictions from the original analysis.

weather conditions are better distinguished by a different set of variables (feature selection) than the vast majority of data records ($\sim 90\%$ of the full data set) associated with less dynamic weather conditions.

Apart from variables wind velocity V10 and PrI, the variable correlation coefficients contributing to the TOB Stage 2 optimized solution for the focused analysis are less than 0.35 (Fig. 11). Nonetheless, the TOB optimization process is able to make good use of these relationships to improve prediction MW accuracy.

The four data records highlighted in Fig. 10 (extreme under-estimates) are characterized by relatively unusual rapidly changing conditions as follows:

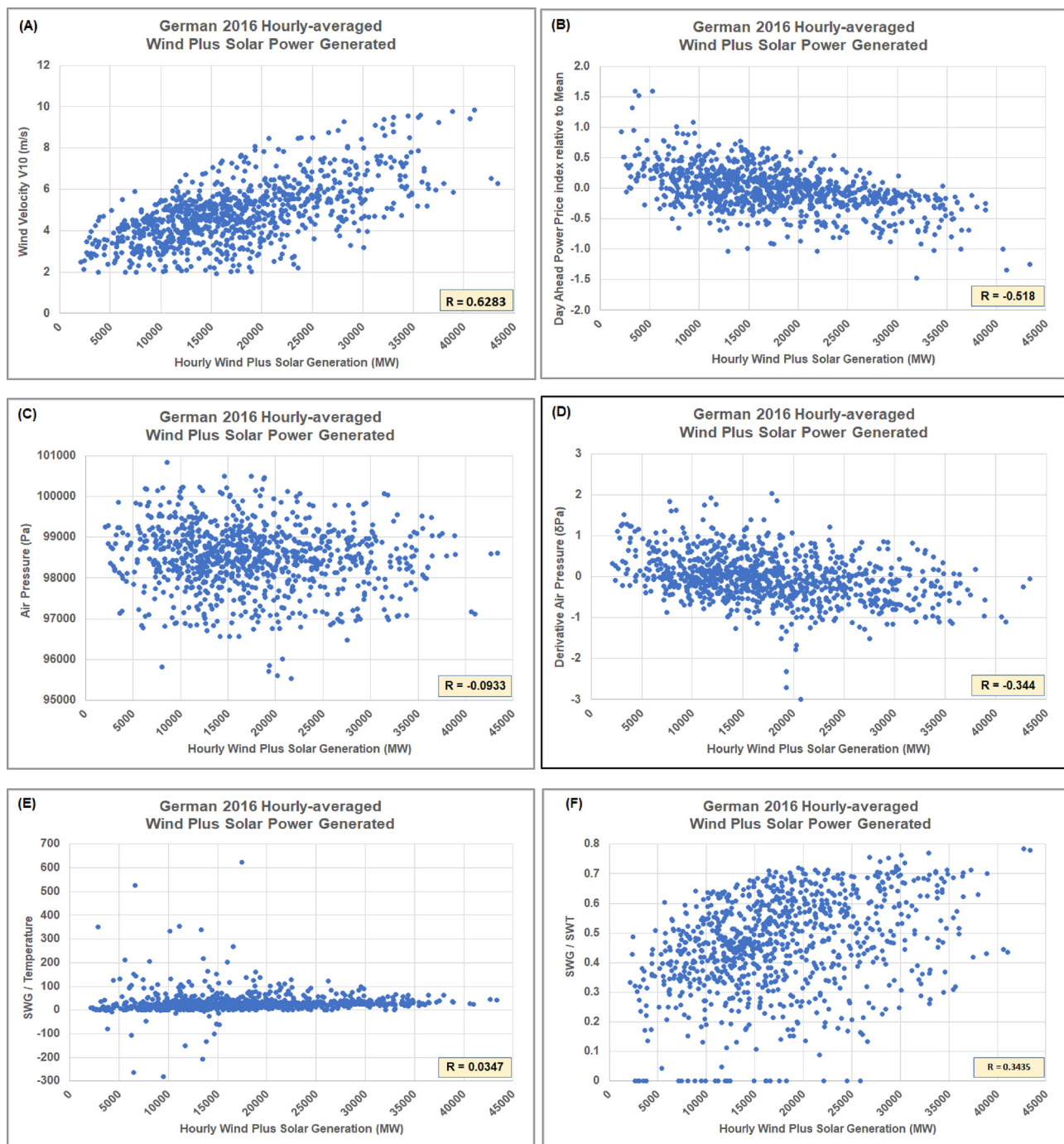


Fig. 11. The six input variables versus power generated for 900 more difficult-to-predict data records selected from the hourly averaged German 2016 dataset (OPSD, 2019; Wood, 2019a) that contribute most to the focused TOB Stage 2 optimum prediction solution (Tables 7 and 8). R is correlation coefficient.

#2827 (27/04/16 18:00) 108 MW solar; 13,911 MW wind; high V50/V10; low SWG/TC; low Pa and high δ Pa

#8068 (02/12/16 03:00) 0 MW solar; 25,854 MW wind; high δ Pa

#6513 (18/09/16 08:00) 10,429 MW solar; 19,762 MW wind; low SWG/ TC; high δ SWG

#3082 (08/05/16 09:00) 24,409 MW solar; 18,936 MW wind; high V50/V10; high SWG/TC; high δ SWG (this is the data record among the difficult-to-predict dataset with the highest combined solar and wind power generated (43,345 MW)).

These four data records represent early morning, late evening or night-time periods for which there are few records with similar conditions in the 900-record dataset. The closest matching records to each of

the identified records have substantially lower MW than the four records identified.

Table 9 provides TOB calculation details for #2827 that provides details of the data record matches involved in the MW prediction of that record. The best matching record with the lowest sum of the squared errors (SumE) is #352 that involves a much lower MW value (more negative normalized MW value, Table 9). Record #352 contributes 37.73 % to the TOB Stage 1 prediction and 45.51 % to the TOB Stage 2 prediction for #2827. The matching record that has the closest MW value to that of #2827 is #184 that contributes 10.29 % to the TOB Stage 1 prediction and 8.89 % to the TOB Stage 2 prediction for #2827. The relatively low contributions of record #184 to the MW predictions

for #2827 relative to data records with much lower MW, lead to the prediction under-estimate for #2827. Neither TOB Stage 1 or Stage 2 (Table 8 solution) MW predictions are good ones for #2827 but the TOB Stage 1 prediction is better because it involves a lower fraction of record #352 and a high fraction of record #184 than the TOB Stage 2 solution. Drilling down into the input variable values of the matching records reveals that for SWG/TC (the dominant variable in the Table 8 solution) there is a closer match between #2827 and #352 than there is between #2827 and #184. Such detailed information helps to assess which of the best matching records most closely reflect the unusual conditions encountered by #2827.

Work is ongoing to establish feature selections that might further improve the MW prediction of data records associated with the most rapidly changing and unusual weather conditions (~1–2% of the full data set). However, it seems likely, for at least some of the extreme prediction outliers identified, further variables that reflect more of the regional variations in this nationally-averaged dataset may be required to provide more precise and reliable MW predictions for these few anomalous data records. Knowing how to identify potentially problematic data records by data mining the full dataset with derivatives and ratios of the available measured data is a valuable step forward towards further improving prediction accuracy of solar plus wind power generation for such compiled datasets.

The findings and applications of this study are not only relevant to country-wide data but are also relevant to compiled data for geographic regions, and more localized cities and large urban areas. Indeed, for sustainable cities of the future intending to rely on renewable energy sources of power supply, it will be essential to provide short-term and long-term power supply forecasts. It will also be necessary for system operators to be prepared for rapidly changing weather conditions that can significantly impact the short-term availability of solar and wind power to sustainable communities relying on it. The TOB algorithm offers potential to assist system operators and regulator in better planning solar and wind power contributions to power supply networks.

6. Conclusions

The country-wide, hourly-averaged dataset of A solar and wind power generation (MW) dataset compiled for Germany in 2016 (on an hourly averaged basis from 256 recording stations countrywide) provides a useful basis for exploring the impact and significance of a set of diverse independent variables on MW predictions. Ten variables covering prevailing weather, environmental conditions and day-ahead electricity prices (Table 3) can be successfully exploited by the transparent open box (TOB) algorithm to provide accurate MW predictions and short-term time series forecasts for the dataset as a whole as well as detailed data mining insight. The best performing TOB Stage 2 optimized solution applied to all 8784 hourly data records achieved a RMSE prediction accuracy of 1174.6 MW and R^2 of 0.9804. It assigned highest weights to the squared errors of surface roughness, air pressure, wind velocity and temperature variables. When that solution was applied to generate one-hour-ahead ($t+1$) forecasts using the previous 1000 h of data it achieved an RMSE prediction accuracy 1631.5 MW and R^2 of 0.9609. For six-hour ahead forecasts, using the previous 1000 h of data prediction accuracy was reduced (RMSE = 2786.4 MW and R^2 = 0.8868). The results indicate that using a longer time period of prior data (e.g. spread over more than one year) could lead to further improvements for TOB's short-term forecasting accuracy.

Data mining the TOB prediction reveals the more extreme prediction outliers. One hundred and twenty-nine hourly periods (some 1.5 percent of all 8784 data records) exceed prediction errors of ± 4472 MW. These records constitute, for the most part, adjacent hours on a few specific days related to atypical, rapidly changing, weather-related conditions. Data mining reveals that these unusual data records are associated with high derivative values (i.e. high rates of

change in solar radiation, wind velocity and/or air pressure) and rapid changes in the ratios between those variables. These derivative and ratio values are used successfully to revise the TOB feature selection to identify, predict and/or filter from the full dataset those data records associated with such unusual and extreme conditions. This leads to improved prediction confidence for the remaining 89.8 % or so of data records (RMSE reduced to 846 MW and R^2 increased to 0.9895). Adjusting feature selection to include derivatives and ratios in a more focused TOB analysis of 900 outlier predictions (~10.2 % of the dataset) improves their prediction accuracy from RMSE = 3544 MW reducing it to 2630 MW (R^2 increases from 0.8027 to 0.8938). Using derivatives of selected variables also has the potential to improve forecasting accuracy, but more work is required on larger datasets (i.e., for multiple years) to confirm that.

Although the use of ratios and derivatives of independent variables to adjust feature selection in a more-focused TOB analysis of the more unusual data records (i.e., those with rapidly changing weather conditions) does significantly improve their MW prediction accuracy overall, a few extreme prediction outliers (1–2% of the dataset) remain. Further work is required on feature selection to identify additional useful independent variables, including their ratios and derivatives, that can further improve prediction and forecasting accuracy (using TOB or other algorithms) for the most rapidly changing and unusual weather conditions in nationally-averaged datasets.

The TOB analysis for the German country-wide dataset for 2016 is also relevant to compiled solar plus wind power compiled datasets focused on regions and large urban areas. Its potential to make prediction and forecasting of renewable power supply more reliable and auditable should contribute to the sustainability of such areas intending to rely upon renewable energy power supplies. It has the potential to do so by improving the ability of grid operators and power supply regulators to predict, forecast and plan for power supply impacts of short-term unexpected weather events.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of conflicting interests

None.

References

- Antonanzas, J., Osorio, N., Escobar, R., Urraca, R., Martinez-de Pison, F. J., & Antonanzas-Torres, F. (2016). Review of photovoltaic power forecasting. *Solar Energy*, 136, 78–111.
- Atkeson, C. G., Moore, A. W., & Schaal, S. (1997). Locally weighted learning. *Artificial Intelligence Review*, 11(1–5), 11–73.
- Birattari, M., Bontempi, G., & Bersini, H. (1999). *Lazy learning meets the recursive least squares algorithm. Advances in neural information processing systems*, 11, Cambridge, MA: MIT Press 375–381.
- Brown, B. G., Katz, R. W., & Murphy, A. H. (1984). Time series models to simulate and forecast wind speed and wind power. *Journal of Climate and Applied Meteorology*, 23(8), 1184–1195.
- Chen, G. H., & Shah, D. (2018). Explaining the Success of Nearest Neighbor Methods in Prediction. *Foundations and Trends® in Machine Learning*, 10(5–6), 337–588.
- Chen, J. L., Liu, H. B., & Wu, W. (2011). Estimation of monthly solar radiation from measured temperatures using support vector machines. *Renewable Energy*, 36(1), 413–420.
- Cover, T. M., & Hart, P. E. (1967). Nearest neighbor pattern classification. *IEEE Transactions on Information Theory*, 13(1), 21–27.
- Energinet (2019). *The Danish national transmission system operator for electricity and natural gas published Elspot Prices*. [accessed 26 March 2019] <https://www.energidataservice.dk/en/dataset/elspotprices>.
- Feng, C., & Zhang, J. (2018). Wind power and ramp forecasting for grid integration. In W. Hu (Ed.). *In book: Advanced wind turbine technology* (pp. 299–315). Springer International Publishing AG. https://doi.org/10.1007/978-3-319-78166-2_11.
- Ferreira, H. (2018). *Predicting wind and solar generation from weather data using Machine Learning*. [accessed 26 March 2019] <https://nbviewer.jupyter.org/github/hugorcf/Renewable-energy-weather/blob/master/renewable.ipynb>.

- Foley, A. M., Leahy, P. G., Marvuglia, A., & McKeogh, E. J. (2012). Current methods and advances in forecasting of wind power generation. *Renewable Energy*, 37, 1–8.
- Frontline Solvers (2019). *Standard excel solver - limitations of nonlinear optimization* (accessed: 26 March 2019) <https://www.solver.com/standard-excel-solver-limitations-nonlinear-optimization>.
- Gensler, A., Henze, J., Sick, B., & Raabe, N. (2016). Deep learning for solar power forecasting — An approach using AutoEncoder and LSTM neural networks. *2016 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* 002858–002865.
- Inman, R. H., Pedro, H. T. C., & Coimbra, C. F. M. (2013). Solar forecasting methods for renewable energy integration. *Progress in Energy and Combustion Science*, 39(6), 535–576.
- Kusiak, A., Zheng, H., & Song, Z. (2009). Short-term prediction of wind farm power: A data mining approach. *IEEE Transactions on Energy Conversion*, 24(1), 125–136.
- Lever, J., Krywinski, M., & Altman, N. (2016). Model selection and overfitting. *Nature Methods*, 13, 703–704. <https://doi.org/10.1038/nmeth.3968> Published online.
- Mangalova, E., & Agafonov, E. (2014). Wind power forecasting using the k-nearest neighbours algorithm. *International Journal of Forecasting*, 30, 402–406. <https://doi.org/10.1016/j.ijforecast.2013.07.008>.
- Mohandes, M. A., Rehmann, S., & Halawani, T. O. (1998). A neural networks approach for wind speed prediction. *Renewable Energy*, 13(3), 345–354.
- OPSD (2019). European power system data in five packages. *Open Power System Data*. [accessed 26 March 2019] <https://data.open-power-system-data.org/>.
- OPSD Time Series (2019). *Load, wind and solar, prices in hourly resolution*. https://doi.org/10.25832/time_series/2018-06-30 [Accessed 26 March 2019].
- OPSD Weather Data (2019). *Hourly geographically aggregated weather data for Europe*. https://doi.org/10.25832/weather_data/2018-09-04 [Accessed 26 March 2019].
- Rana, M., Koprinska, I., & Agelidis, V. G. (2016). Solar power forecasting using weather type clustering and ensembles of neural networks. *International Joint Conference on Neural Networks (IJCNN)* (pp. 4962–4969).
- Sivaneasan, B., Yu, C. Y., & Goh, K. P. (2017). Solar forecasting using ANN with fuzzy logic pre-processing. *Energy Procedia*, 143, 727–732.
- Stetco, A., Dinmohammadi, F., & Zhao, X. (2019). Machine learning methods for wind turbine condition monitoring: A review. *Renewable Energy*, 133, 620–635.
- Treiber, N. A., Heinemann, J., & Kramer, O. (2016). Wind power prediction with machine learning. in J. Lassig et al. *Computational Sustainability, Studies in Computational Intelligence*, 645, 13–29. https://doi.org/10.1007/978-3-319-31858-5_2.
- Vladislavleva, E., Friedrich, T., Neumann, F., & Wagner, M. (2013). Predicting the energy output of wind farms based on weather data: Important variables and their correlation. *Renewable Energy*, 50, 236–243. <https://doi.org/10.1016/j.renene.2012.06.036>.
- Voyant, C., Nottin, G., Kalogirou, S., Nivet, M. L., Paoli, C., Motte, F., et al. (2017). Machine Learning methods for solar radiation forecasting: A review. *Renewable Energy*, 105, 569–582.
- Wang, J., Li, P., Ran, R., Che, Y., & Zhou, Y. (2018). A short-term photovoltaic power prediction model based on the gradient boost decision tree. *Applied Science*, 8, 689. <https://doi.org/10.3390/app8050689>.
- Wood, D. A. (2018a). A transparent Open-Box learning network provides insight to complex systems and a performance benchmark for more-opaque machine learning algorithms. *Advances in Geo-energy Research*, 2(2), 148–162.
- Wood, D. A. (2019a). German country-wide renewable power generation from solar plus wind mined with an optimized data matching algorithm utilizing diverse variables. *Energy Systems*. <https://doi.org/10.1007/s12667-019-00347-x> in press (accepted July 2019).
- Wood, D. A. (2018b). Transparent open-box learning network provides auditable predictions for coal gross calorific value. *Modeling Earth Systems and Environment*. <https://doi.org/10.1007/s40808-018-0543-9> Published online 16 November, 2018.
- Wood, D. A. (2019b). Sensitivity analysis and optimization capabilities of the transparent open-box learning network in predicting coal gross calorific value from underlying compositional variables. *Modeling Earth Systems and Environment*. <https://doi.org/10.1007/s40808-019-00583-1> Published online 21 March, 2019.
- Wood, D. A. (2019c). German solar power generation data mining and prediction with transparent open box learning network integrating weather, environmental and market variables. *Energy Conversion and Management*, 196, 354–369.
- Yang, X.-S., & He, X. (2013). Firefly algorithm: Recent advances and applications. *Int. J. Swarm Intelligence*, 1(1), 36–50.